

PERSONAL HOMELAB INFRASTRUCTURE

merox.dev Infrastructure Overview

Kubernetes · GitOps · Self-Hosted Cloud

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June 2026 · v2.0

3

K8S
NODES

30+

SERVICES

100%

GITOPS

~€0

CLOUD
COST



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1. Executive Summary

This document provides a complete reference for the **merox.dev homelab infrastructure** — a production-grade self-hosted environment running on commodity hardware and free-tier cloud services. The design philosophy is: declarative, reproducible, and zero-cost cloud.



Kubernetes Cluster

3-node Talos Linux HA cluster (no workers — all control-plane) running on Proxmox VMs. Managed 100% via FluxCD GitOps. A single `git push` deploys or updates any workload.



Oracle Cloud VPS

Always-free ARM VPS (4 vCPU / 24GB) hosting off-site services: reverse proxy, SSO, DNS, S3 backup storage, monitoring, and AI agent runtime.



Zero Open Ports

All external traffic routes through **Cloudflare Tunnel** (no firewall holes). Internal management uses **Tailscale** mesh VPN. No public IP required.



AI Automation

OpenClaw AI agent framework runs Claude-powered agents — manage infrastructure, publish blog posts, and get daily briefings via Telegram.

KEY FACTS

Git repo: [meroxdotdev/infrastructure](#) · Secrets: SOPS/AGE encrypted · Rebuild time from scratch: ~50 min · Monthly cloud cost: ~€0 (Oracle free tier) · Fallback VPS on-demand: ~€5.39/mo (Hetzner)

2. High-Level Architecture & Network Topology

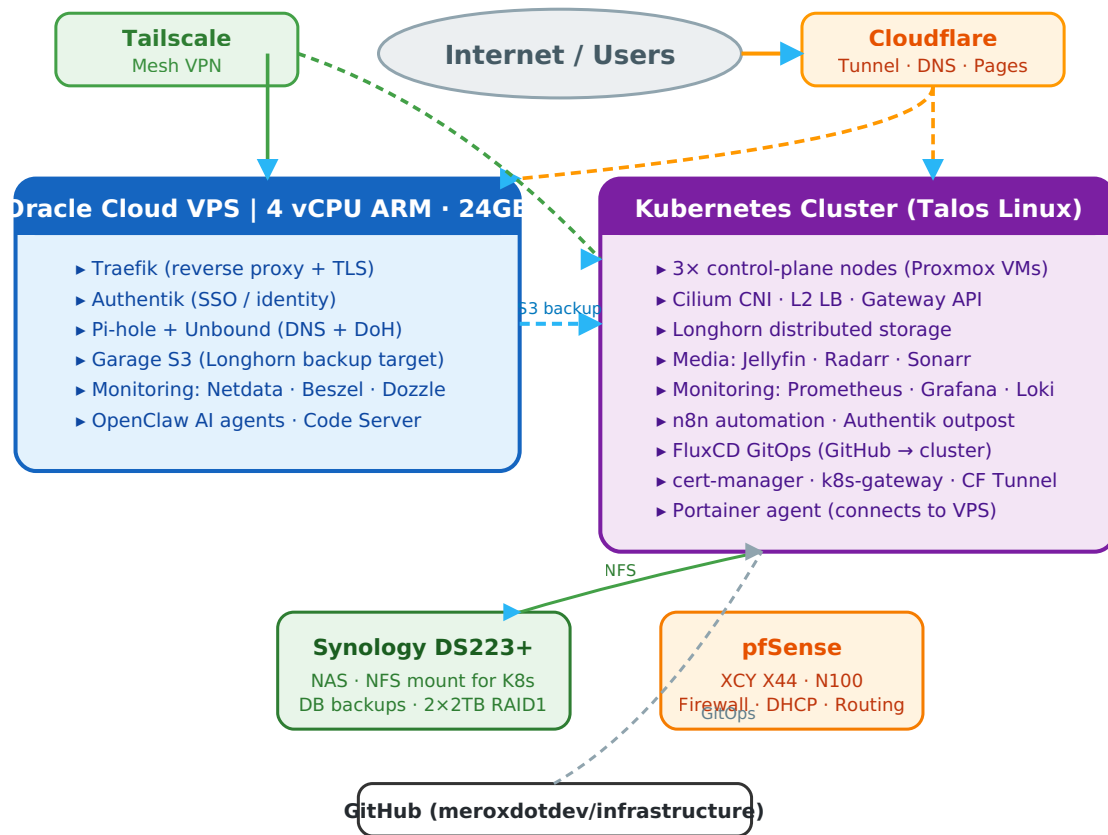


Figure 1 — High-level infrastructure topology showing all major components and their interconnections

3. Hardware Inventory

Dell OptiPlex 3050 #1

K8s Node · Proxmox VM

CPU: Intel i5-6500T (4C/4T)
RAM: 16 GB DDR4
Storage: 128 GB NVMe
IP: 10.57.57.80

Dell OptiPlex 3050 #2

K8s Node · Proxmox VM

CPU: Intel i5-6500T (4C/4T)
RAM: 16 GB DDR4
Storage: 128 GB NVMe
IP: 10.57.57.82

Beelink GTi 13 Pro

K8s Node · Proxmox VM

CPU: Intel i9-13900H (14C/20T)
RAM: 64 GB DDR5
Storage: 2 × 2 TB NVMe
IP: 10.57.57.84 · iGPU: Intel Arc (i915)

Dell PowerEdge R720

Proxmox Backup Server

CPU: 2 × Xeon E5-2697v2 (24C/48T)
RAM: 192 GB DDR3
Storage: Large HDD pool
Role: PBS + DR VMs

Synology DS223+

NAS · NFS + Backup

Storage: 2 × 2 TB HDD RAID1
Protocol: NFS (K8s volumes)
IP: 10.57.57.201
Role: DB backups + media storage

XCY X44

Firewall / Router

CPU: Intel N100
RAM: 8 GB
OS: pfSense
Role: Firewall · DHCP · Routing

Oracle Cloud ARM VPS

Off-site Cloud (Free tier)

CPU: 4 × vCPU ARM Ampere A1
RAM: 24 GB
Storage: 200 GB block storage
Cost: €0/mo (always free)

Network Addressing

Device/Service	IP / Address	Notes
K8s VIP (Talos)	10.57.57.88	Kubernetes API endpoint (HA)
K8s Node 1	10.57.57.80	OptiPlex 3050 #1
K8s Node 2	10.57.57.82	OptiPlex 3050 #2
K8s Node 3	10.57.57.84	Beelink GTi 13 Pro
Synology NAS	10.57.57.201	NFS + DB backup target
qBittorrent pod	10.57.57.102	Fixed IP (Cilium)
Portainer agent pod	10.57.57.103	Fixed IP (Cilium)
Oracle VPS (Tailscale)	100.72.22.38	Management access

4. Oracle Cloud VPS — Off-site Services

All services run as Docker Compose stacks under `/srv/docker/oracle-cloud/`. Traefik handles TLS termination and routing. Authentik provides SSO for protected services.

Service	URL / Access	Purpose	Stack
Traefik	traefik.cloud.merlox.dev	Reverse proxy · ACME TLS (Let's Encrypt) · routes all VPS services	traefik/
Authentik	sso.merlox.dev	SSO identity provider — Google-only login, no new signups	authentik/
Pi-hole + Unbound	pihole.cloud.merlox.dev/admin	DNS ad-blocking + DNS-over-HTTPS recursive resolver	docker-compose.yml
Portainer EE	100.72.22.38:9000 (Tailscale)	Container management UI — manages all Docker services on VPS	docker-compose.yml
Homepage	inside.merlox.dev (Tailscale)	Internal dashboard showing K8s + Proxmox + router status	docker-compose.yml
Joplin Server	joplin.cloud.merlox.dev	Self-hosted notes sync with PostgreSQL backend	docker-compose.yml
Uptime Kuma	status.merlox.dev	Service uptime monitoring + alerting	uptime-kuma/
Guacamole	rmt.merlox.dev	Browser-based remote desktop gateway (RDP/VNC/SSH) · Authentik SSO	guacamole/
Garage S3	garage.cloud.merlox.dev	S3-compatible object storage — receives Longhorn volume backups from homelab	garage/
Netdata	netdata.cloud.merlox.dev	Real-time metrics — parent node + 3 K8s child nodes	docker-compose.yml
Beszel	beszel.cloud.merlox.dev	Lightweight host monitoring dashboard	beszel/
Dozzle	dozzle.cloud.merlox.dev	Real-time Docker log aggregation UI	docker-compose.yml
Glances	glances.cloud.merlox.dev	System resource monitoring (CPU, RAM, disk, network)	docker-compose.yml
OpenClaw Dashboard	agents.cloud.merlox.dev	AI agent control panel — updated nightly	agents-dashboard/
Code Server	code.cloud.merlox.dev	Browser-based VS Code for remote editing	docker-compose.yml
Filebrowser	files.cloud.merlox.dev	Web-based file manager for VPS storage	filebrowser/

VPS Management Commands

```
cd /srv/docker/oracle-cloud
```

```
make health-check    # verify all services running
make setup            # full idempotent redeploy via Ansible
make update           # OS package updates only
make check            # dry-run (--check --diff)
make restore          # interactive restore wizard (Joplin / Authentik)
make cleanup          # remove unused Docker images/volumes
make dr-full          # provision fallback VPS on Hetzner (~15 min)
```

5. Kubernetes Cluster — On-premise

3-node **Talos Linux** cluster — all nodes run both control-plane and workloads (no dedicated workers).
Managed via **FluxCD v2** GitOps. Charts sourced via OCI where possible (bjw-s-labs app-template 4.x).

Infrastructure Services

Service	Namespace	Role
Cilium	kube-system	CNI · Gateway API · L2 LoadBalancer · NetworkPolicy
Longhorn	longhorn-system	Distributed block storage · S3 backup to Garage
cert-manager	cert-manager	Automated TLS (Let's Encrypt ACME)
Cloudflare Tunnel	network	External access — zero open ports required
k8s-gateway	network	Internal DNS for *.merox.dev
external-dns	network	Syncs K8s ingress → Cloudflare DNS
FluxCD	flux-system	GitOps controller (GitHub → cluster)
intel-device-plugin	kube-system	Exposes i915 iGPU for Jellyfin transcoding

Observability Stack

Service	Namespace	Role
Prometheus	observability	Metrics collection + alerting rules
Grafana	observability	Dashboards (K8s, Longhorn, apps)
Loki	observability	Log aggregation backend
Promtail	observability	Log shipper (DaemonSet)
AlertManager	observability	Alerts + healthchecks.io heartbeat
Portainer Agent	default	Connects to Portainer EE on VPS

Application Workloads

Service	Namespace	Purpose	Notes
Jellyfin	default	Media server — movies, TV, music	Intel i915 iGPU hardware transcoding · NodePort 30096 (Samsung TV)
Jellyseerr	default	Media request management UI	Users request content → Radarr/Sonarr pick it up
Radarr	default	Movie automation & library management	Monitors NZB/torrent indexers, downloads, renames
Sonarr	default	TV show automation	Same as Radarr but for TV series
Prowlarr	default	Unified torrent/NZB indexer	Single config point, syncs to Radarr/Sonarr
qBittorrent	default	Torrent download client	Fixed IP: 10.57.57.102 (Cilium)
n8n	default	Workflow automation (no-code)	Integrations: GitHub, webhooks, notifications
Authentik outpost	default	SSO proxy for K8s-hosted apps	Connects to Authentik instance on VPS

Key Cluster Commands

```
kubectl get nodes                # check node status
kubectl get pods -A | grep -v Running # find problem pods
kubectl get helmreleases -A       # all Helm releases + status
task reconcile                    # force Flux sync

# Node operations
task talos:apply-node IP=10.57.57.80 # apply Talos config
task talos:upgrade-node IP=10.57.57.80 # upgrade Talos version
task talos:upgrade-k8s              # upgrade Kubernetes
```

6. Networking & Access Strategy

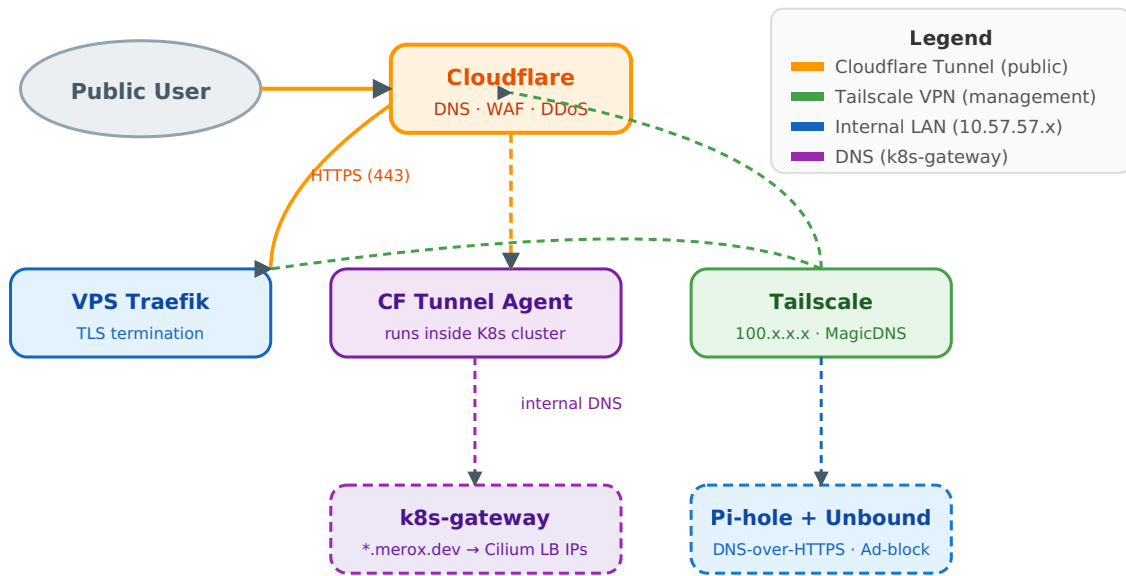


Figure 2 — Network access paths: public traffic via Cloudflare Tunnel, management via Tailscale VPN

Cloudflare Tunnel (Public Access)

- Zero open firewall ports on VPS or homelab
- All `*.merox.dev` public services route through tunnel
- TLS terminated at Cloudflare edge (free)
- DDoS protection & WAF included
- K8s tunnel agent deployed as pod in cluster
- VPS tunnel runs via Traefik container

Tailscale (Management VPN)

- All nodes (VPS + K8s + Proxmox) in same Tailscale network
- MagicDNS for hostname resolution
- VPS acts as exit node
- Portainer EE & Homepage accessible only via Tailscale
- Longhorn → Garage S3 backups travel over Tailscale
- Prevents accidental public exposure of mgmt interfaces

Internal DNS (k8s-gateway)

The `k8s-gateway` pod resolves `*.merox.dev` internally to Cilium L2 LoadBalancer IPs, so internal clients use the same URLs as external ones but traffic stays on-LAN (split-horizon DNS).



7. Storage & Backup Architecture

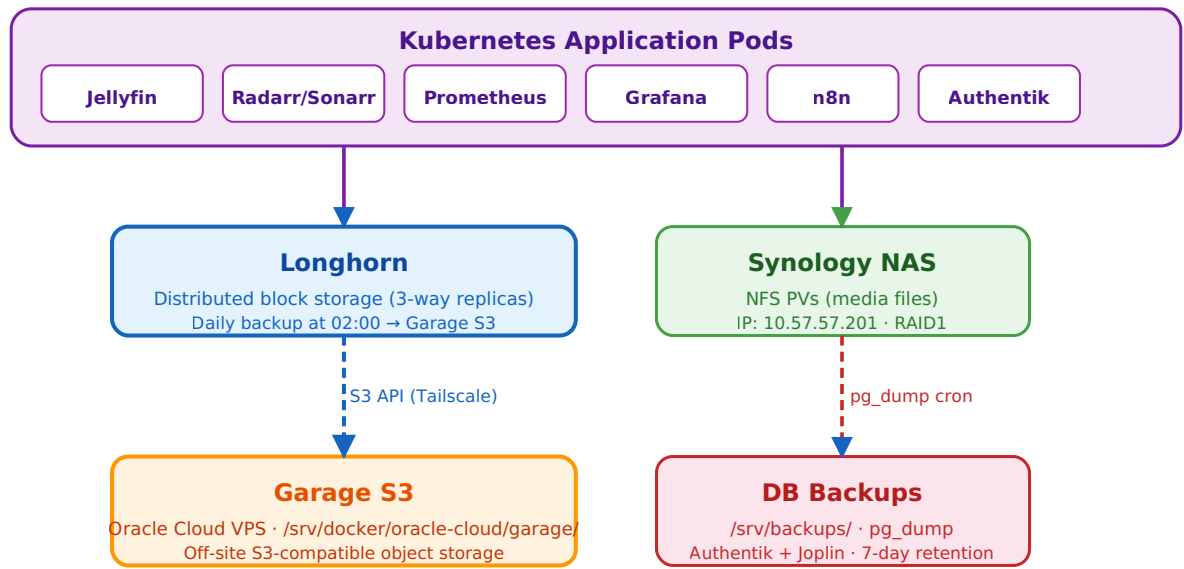


Figure 3 — Storage layers and backup data flows (Longhorn → Garage S3, databases → /srv/backups)

Storage Layers

Layer	Technology	Used For
Block (replicated)	Longhorn	App config, databases, state
File (shared NFS)	Synology DS223+	Media files, large datasets
Object (S3)	Garage (VPS)	Longhorn snapshot backups

Backup Schedule

What	When	Where
Longhorn volumes	Daily 02:00	Garage S3 (off-site VPS)
Authentik PostgreSQL	Manual (pre-change)	/srv/backups/authentik/
Joplin PostgreSQL	Automatic cron	/srv/backups/

CRITICAL — BACK UP AGE.KEY SEPARATELY

`/srv/kubernetes/infrastructure/age.key` is the SOPS master decryption key. If lost, all encrypted K8s secrets (Cloudflare token, Authentik secrets, Longhorn S3 credentials) become unrecoverable. Keep an offline copy.

8. GitOps & Automation

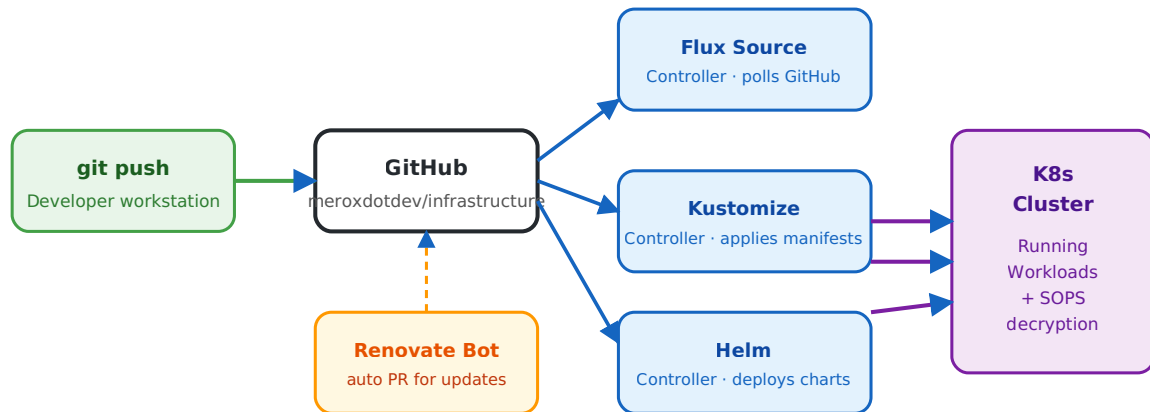


Figure 4 — FluxCD GitOps pipeline: git push triggers Flux reconciliation, Renovate automates dependency updates

FluxCD Pipeline

- Source Controller polls GitHub every 1 min
- Kustomize Controller applies `kubernetes/` manifests
- Helm Controller reconciles all HelmRelease resources
- SOPS/AGE decrypts secrets at apply time (age.key in-cluster)
- Drift detection: Flux auto-reverts manual kubectl changes
- Force sync: `task reconcile`

Renovate Automation

- Runs every weekend via GitHub Actions
- Opens PRs for: Helm chart updates, container image tags
- Also tracks: Talos + Kubernetes versions (`.mise.toml`)
- Config: `.renovaterc.json5`
- Tags annotated with `# renovate:` in manifests
- VPS Ansible also updated via Renovate

Secrets Management (SOPS/AGE)

```
sops kubernetes/apps/<namespace>/<app>/app/secret.sops.yaml # edit secret
find . -name "*.sops.*" -exec sops updatekeys {} \;          # rotate AGE key
```

9. AI Agents — OpenClaw

OpenClaw is an AI agent framework running Claude-powered agents on the VPS. Agents are triggered via **Telegram bot** or internal cron schedules. The `infra` agent can run `kubectl` and `docker` commands remotely.

Agent	Trigger	Purpose
news	Cron + Telegram	Daily briefing — HackerNews + RSS feeds, summarised by Claude
blog	Telegram	Writes and publishes posts to merox.dev via GitHub Actions
infra	Telegram	Runs <code>kubectl</code> / <code>docker</code> commands — natural language infra management
costs	Telegram	Infrastructure cost tracking and reporting
design	Telegram	Visual content generation
orchestrator	Internal	Routes messages between agents + handles scheduled tasks
dashboard	Internal cron	Updates agents.cloud.merox.dev nightly
renovate	Internal	Git dependency sync + PR creation

```
# Agent status & management
XDG_RUNTIME_DIR=/run/user/$(id -u openclaw) \
  sudo -u openclaw systemctl --user status openclaw-gateway

sudo -u openclaw journalctl --user -u openclaw-gateway -f
tail -f /home/openclaw/.openclaw/logs/*.log 2>/dev/null
```

10. Security Model

Secret Handling

Secret Type	Method
K8s secrets	SOPS/AGE encrypted <code>*.sops.yaml</code>
VPS Ansible secrets	Ansible Vault (<code>vps/</code>)
Docker env vars	<code>.env</code> file (gitignored)
Agent API keys	<code>.openclaw/.env</code> (gitignored)
Talos bootstrap	<code>talsecret.sops.yaml</code> (SOPS)

Access Control

Layer	Mechanism
External HTTP	Cloudflare Tunnel (no open ports)
Management	Tailscale mesh VPN
SSO	Authentik (Google-only, no signups)
K8s RBAC	Talos native (hardened by default)
Network policy	Cilium NetworkPolicy

11. Disaster Recovery

PREREQUISITES (HAVE THESE READY BEFORE ANY DR SCENARIO)

`age.key` (SOPS master key) · Ansible Vault password · Valid Tailscale auth key · Cloudflare API token

DR Scenario Matrix

Scenario	Action	RTO
K8s cluster lost (nodes dead)	DR.md — provision DR VMs, bootstrap Talos, restore Longhorn from S3	~35 min
VPS lost (Oracle reclaims free tier)	<code>cd vps && make dr-full</code> → <code>make restore</code>	~15 min
Full rebuild from scratch	DEPLOY.md: Phase 1 (VPS) → Phase 2 (K8s) → Phase 3 (Agent)	~50 min
Single node lost	Provision replacement VM in Proxmox, <code>task talos:apply-node</code>	~10 min
New hardware (different IPs)	Edit <code>talconfig.yaml</code> , <code>cluster-vars.yaml</code> , <code>cilium/networks.yaml</code>	~20 min
HelmRelease stuck	<code>flux suspend/resume helmrelease <name></code>	<5 min
VPS only (K8s OK)	<code>make dr-full</code> on any machine with the repo + vault pass	~15 min

Full Rebuild Steps (Phase 1 → 3)

- Phase 1 — VPS (~15 min)**
`cd vps && make dr-full` — provisions Hetzner fallback VPS + deploys all Docker services via cloud-init.
Then: `make restore` to restore Joplin + Authentik databases from backups.
- Phase 2 — Kubernetes (~20 min)**
Copy `age.key` to repo. Edit `talos/talconfig.yaml` (node IPs, install disk) and `kubernetes/components/common/cluster-vars.yaml`.
Run: `task bootstrap:talos`
- Phase 3 — AI Agents (~15 min)**
Create `openclaw` user, run `claude login`, run `openclaw onboard`, fill in `.env`, enable systemd service.
- Validation**
`kubectl get nodes` → all Ready · `docker ps | wc -l` → ~16 containers · Test Telegram bot reply.

Troubleshooting Quick Reference

```
# Flux not reconciling
flux get sources git -A
flux logs --level=error
flux reconcile kustomization cluster-apps --with-source

# HelmRelease stuck
kubectl get helmreleases -A | grep -v True
flux suspend helmrelease <name> -n <namespace>
flux resume  helmrelease <name> -n <namespace>

# Pod issues
kubectl -n <ns> describe pod <pod>
kubectl -n <ns> logs <pod> --previous

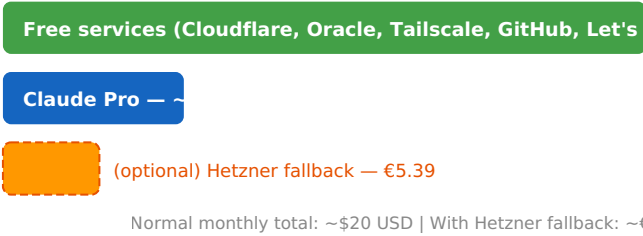
# Node unreachable
talosctl -n <node-ip> health
talosctl -n <node-ip> dmesg
```



12. External Dependencies & Costs

Service	Purpose	Cost	Tier
Cloudflare	DNS · CDN · Tunnel · WAF · Pages (blog CI/CD)	€0/mo	Free
Tailscale	Management VPN mesh (all nodes + VPS)	€0/mo	Free
Oracle Cloud	Primary VPS (4 vCPU ARM, 24GB RAM, 200GB)	€0/mo	Always Free
Hetzner VPS	Fallback VPS — provision only if Oracle free tier lost	€5.39/mo	On-demand
Anthropic / Claude	AI model powering OpenClaw agents (Claude Pro OAuth)	~\$20/mo	Claude Pro
GitHub	Repos + GitHub Actions (CI for blog, Renovate)	€0/mo	Free
Let's Encrypt	HTTPS certificates (auto-renew via cert-manager)	€0/mo	Free
Proxmox	Hypervisor for K8s VMs (runs on own hardware)	€0/mo	Own HW
Synology DS223+	NAS — NFS for K8s + DB backup storage	€0/mo	Own HW

Monthly Infrastructure Cost Breakdown



Repository Map

What	GitHub Repo	Local Path
K8s Flux manifests + Talos config	meroxdotdev/infrastructure	/srv/kubernetes/infrastructure/
Ansible + Terraform VPS DR	meroxdotdev/infrastructure	/srv/kubernetes/infrastructure/vps/
Docker Compose VPS services	meroxdotdev/cloudlab-merox	/srv/docker/oracle-cloud/
OpenClaw agent config + infra skill	meroxdotdev/infrastructure	/srv/kubernetes/infrastructure/agent/
Blog (Astro)	meroxdotdev/merox (private)	/srv/merox/

13. Day-to-Day Operations Quick Reference

Kubernetes Cluster

```
kubectl get nodes
kubectl get pods -A | grep -v Running
kubectl get helmreleases -A
kubectl get kustomizations -A
cilium status

task reconcile # force Flux sync

# Talos node ops
task talos:generate-config
task talos:apply-node IP=10.57.57.80
task talos:upgrade-node IP=10.57.57.80
task talos:upgrade-k8s
```

VPS / Docker Services

```
cd /srv/docker/oracle-cloud

docker ps
docker ps | grep -v Up
docker logs <name> -f
docker compose up -d

cd vps/
make health-check
make setup
make update
make restore
make dr-full
```

Flux Troubleshooting

```
flux get sources git -A
flux get kustomizations -A
flux logs --level=error
flux reconcile kustomization \
  cluster-apps --with-source

# HelmRelease stuck
flux suspend helmrelease <name> -n <ns>
flux resume helmrelease <name> -n <ns>
```

Node Maintenance

```
# Drain node before maintenance
kubectl drain <node> \
  --ignore-daemonsets \
  --delete-emptydir-data

# After maintenance
kubectl uncordon <node>

# Wait 1-2h between disk swaps
# for Longhorn replica rebuild
```

Longhorn Storage

```
kubectl -n longhorn-system get volumes
kubectl -n longhorn-system get nodes.longhorn.io

# Remove orphaned replicas
kubectl get orphan -n longhorn-system -o name | \
  xargs kubectl delete -n longhorn-system
```

Garage S3

```
docker exec garage /garage status
docker exec garage /garage bucket list
kubectl -n longhorn-system \
  get secret minio-secret
```

SOPS Secrets

```
sops kubernetes/apps/<ns>/<app>/\
  app/secret.sops.yaml

# After AGE key rotation
find . -name "*.sops.*" \
  -exec sops updatekeys {} \;
```

FULL DOCUMENTATION

Complete index: [/srv/kubernetes/infrastructure/README.md](#) · Full rebuild guide: [DEPLOY.md](#) · K8s DR procedure: [DR.md](#) · Jellyfin post-restore: [docs/jellyfin-post-restore.md](#)

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